

1. Consider the function $h: \mathbb{R} \rightarrow \mathbb{R}$, where $h(t) = (f \circ g)(t) = f(g(t))$ with

5 points

$$g(t) = \mathbf{x} = \begin{bmatrix} t \cos t \\ t \sin t \end{bmatrix}, \quad t \in \mathbb{R}$$

$$f(\mathbf{x}) = \exp(x_1 x_2^2), \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in \mathbb{R}^2$$

$$\frac{dh}{dt} = \exp(x_1 x_2^2) [x_2^2 (\cos t - t \sin t) + 2x_1 x_2 (\sin t + t \cos t)] \text{ with}$$

$$x_1 = t \cos t, \quad x_2 = t \sin t$$

$$\frac{df}{dx} = [x_2^2 \exp(x_1 x_2^2) \quad 2x_1 x_2 \exp(x_1 x_2^2)]$$

$$\frac{dh}{dt} = \frac{df}{dg} \frac{dg}{dt}$$

$$\frac{dg}{dt} = \begin{bmatrix} \sin t - t \cos t \\ \cos t + t \sin t \end{bmatrix}$$

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$$\frac{df}{dx} = [x_1 x_2^2 \quad 2x_1 x_2 x_2^2]$$

$$\frac{dh}{dt} = \cos t - t \sin t - 2t \sin t (\sin t + t \cos t)$$

2. Compute $\frac{d^2 f}{dx^2}$ of the following function using the chain rule.

1 point

$$a = x^2$$

$$b = \exp(a)$$

$$c = a + b$$

$$d = \log(c)$$

$$e = \sin(c)$$

$$f = d + e$$

$$\frac{df}{dx} = \frac{(1 + \cos(x^2 + \exp(x^2)))(x^2 + \exp(x^2))(2x + 2x \exp(x^2))}{x^2 + \exp(x^2) + \log(x^2)}$$

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3. What is $\frac{df}{dx}$ where

1 point

$$f = \cos(t^2)$$

$$t = x^3$$

$$\textcircled{a} -6x^3 \sin(x^6)$$